Package ‘GR2MSemiDistr’

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Type Package

Title Semi-Distributed GR2M Model for Catchment to Large-Scale Hydrological Applications

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Description Implements a semi-distributed adaptation of the GR2M monthly water balance model with spatial discretization into subbasins and routing based on a transfer matrix interpreted as a directed graph. The model supports both small catchments and large-sample applications with thousands of subbasins, using regionalized parameters and subbasin-level forcings to simulate precipitation, evapotranspiration, runoff, and discharge. It includes warm-up periods, initial states, saving options, regional parameter correction, and performance evaluation through multiple objective functions. Seamlessly integrates with the 'airGR' hydrological modeling framework and the 'terra' spatial ecosystem, making it suitable for research, operational forecasting, and water resources assessment. Future versions will include multi-objective calibration methods, bias correction of gridded forcings, automated preparation of inputs from observations, and advanced routing based on linear reservoir schemes.

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Depends R (>= 4.0)

Imports airGR, hydroGOF, terra, lubridate, tictoc, sf, rtop, exactextractr, Matrix, igraph

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Contents

`Build_Transfer_Matrix` *Build transfer matrix for subbasins connectivity*

Description

Create a dense 0/1 transfer matrix (as a data.frame) from a river reach/subbasin layer that contains a unique ID (‘COMID‘) and its immediate downstream ID (‘NextDownID‘). **Rows are receivers**, **columns are donors**. The result is a square (nsub × nsub) matrix of subbasin connectivity whose row and column names must match ‘COMID‘.

Usage

```
Build_Transfer_Matrix(Rivers)
```

Arguments

`Rivers` SpatVector. River reaches or subbasins. Must be a vector object from the ‘terra‘ package. Its attribute table must include:

COMID Unique subbasin/reach identifier (character or numeric, internally coerced to character).

NextDownID Identifier of the immediate downstream subbasin/reach. Outlets (no downstream) must be coded with -1.

Geometry is ignored; only the attribute table is used to build the connectivity.

Details

Designed to plug into routing steps used by ‘Run_GR2MSemiDistr()‘ and related functions in the package.

Value

dgCMatrix (sparse matrix). Subbasin connectivity matrix where rows represent receiving (downstream) subbasins and columns represent donor (upstream) subbasins. Row and column names are set to ‘COMID‘.

`Create_Forcing_Inputs` *Extract and prepare model input data in the airGR format (DatesR, P, E, and optionally Q) using gridded monthly precipitation and evapotranspiration.*

Description

This function extracts spatially averaged monthly precipitation and potential evapotranspiration values for each subbasin, and optionally includes observed streamflow data, to produce a formatted data frame compatible with the airGR hydrological modeling framework.

Usage

```
Create_Forcing_Inputs(
  Subbasins,
  Precip,
  PotEvap,
  Qobs = NULL,
  DateIni,
  DateEnd,
  IniGrids = "01/1981",
  Save = FALSE,
  Update = FALSE,
  Members = FALSE
)
```

Arguments

`Subbasins`	SpatVector. Subbasin geometries. Must include attributes: 'COMID' (unique subbasin identifier) and 'Region' (character code for the hydro-climatic region).
`Precip`	SpatRaster. Monthly precipitation fields in [mm/month]. Dimensions and time coverage must include at least the requested period ('DateIni'–'DateEnd').
`PotEvap`	SpatRaster. Monthly potential evapotranspiration fields in [mm/month]. Must align with the precipitation raster in space and time.
`Qobs`	numeric vector (optional). Observed streamflow series at the basin outlet, in [m^3/s]. Length must match the number of simulated months if provided. Default is 'NULL'.
`DateIni`	character. Start date of the output time series in '"mm/yyyy"' format.
`DateEnd`	character. End date of the output time series in '"mm/yyyy"' format.
`IniGrids`	character. Initial date of the gridded precipitation and evapotranspiration datasets in '"mm/yyyy"' format. Used to align raster time indices with the requested simulation period.
`Save`	logical. If 'TRUE', saves the resulting data frame to a tab-separated '.txt' file inside the './Inputs' directory. Default is 'FALSE'.
`Update`	logical. If 'TRUE', returns only the last month of data (useful for operational forecasting/updates). Default is 'FALSE'.
`Members`	logical. If 'TRUE', the number of ensemble members is inferred from the raster layers (stacked as members × months). If 'FALSE', a single deterministic series is processed. Default is 'FALSE'.

Value

data.frame with model inputs in the airGR format, including the following columns:

DatesR POSIXct. Monthly dates covering the simulation period.

P_x numeric. Mean precipitation for subbasin x [mm/month], where x is the COMID.

E_x numeric. Mean potential evapotranspiration for subbasin x [mm/month], where x is the COMID.

Q numeric (optional). Observed streamflow at the basin outlet [m³/s], included only if ‘Qobs‘ was provided.

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). Water, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Define simulation period and grid start date
start_date <- "01/1981"
end_date   <- "12/2016"
ini_grids  <- "01/1981"

# Create input data for the model
model_inputs <- Create_Forcing_Inputs(
  Subbasins = cat,
  Precip = grid_pr,
  PotEvap = grid_pe,
  Qobs = qobs,
  DateIni = start_date,
  DateEnd = end_date,
  IniGrids = ini_grids,
)

# Select a target COMID to plot
target_comid <- "10"

# Plot precipitation for the target subbasin
P_col <- paste0("P_", target_comid)
plot(model_inputs$DatesR, model_inputs[[P_col]], type = "h", lwd = 2, col = "dodgerblue",
    ylab = "Precipitation [mm/month]", xlab = "Date",
    main = sprintf("Precipitation in Subbasin %s", target_comid))

# Plot potential evapotranspiration for the target subbasin
E_col <- paste0("E_", target_comid)
plot(model_inputs$DatesR, model_inputs[[E_col]], type = "o", lwd = 2, col = "darkgreen",
     ylab = "PET [mm/month]", xlab = "Date",
     main = sprintf("Potential Evapotranspiration in Subbasin %s", target_comid))

# Plot observed streamflow at outlet (if available)
if ("Q" %in% names(model_inputs)) {
  plot(model_inputs$DatesR, model_inputs$Q, type = "l", col = "darkblue", lwd = 1.5,
       ylab = "Q [m³/s]", xlab = "Date",
```

```
        main = "Observed Streamflow at Basin Outlet")
}
```

`Load_example_data` *Load example hydrological data included in the package*

Description

Loads spatial and tabular example data (raster and vector) from 'inst/extdata/'.

Usage

```
Load_example_data()
```

Value

A list with objects: 'cat', 'dem', 'grid_pr', 'grid_pe', 'qobs'

Examples

```
# Load example data

data <- GR2MSemiDistr::Load_example_data()
names(data)

cat     <- data$cat       # Subbasin boundaries (SpatVector)
dem     <- data$dem       # DEM (SpatRaster)
qobs    <- data$qobs      # Observed streamflow (data.frame)
grid_pr <- data$grid_pr   # Precipitation (SpatRaster)
grid_pe <- data$grid_pe   # Potential evapotranspiration (SpatRaster)
matrixT <- data$matrixT   # Connectivity matrix (data.frame)

terra::plot(mask(dem, cat), alpha = 0.8, main = "Subbasins on DEM")
terra::plot(cat, add = TRUE)
terra::text(cat, labels = cat$COMID, cex = 0.6)
```

`Optim_GR2MSemiDistr` *Parameter optimization for a semi-distributed GR2M model using the SCE-UA algorithm.*

Description

Optimizes the GR2M monthly water balance model parameters (X1, X2, fp, fe) for multiple calibration regions using the SCE-UA algorithm. The model is run in a semi-distributed setup over multiple subbasins grouped by region. Users can select from various objective functions, including composite metrics. Parameter bounds must be provided, and regions can be excluded from calibration.

Usage

```
Optim_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  TransferMatrix,
  Outlet,
  Parameters.Min = c(100, 0.1, 0.8, 0.8),
  Parameters.Max = c(2000, 10, 1.2, 1.2),
  Optimization = "OF10",
  Max.Functions = 1000,
  WarmUp = 36,
  No.Optim = NULL,
  w1 = 0.6,
  w2 = 0.3,
  w3 = 0.2
)
```

Arguments

`Data` data.frame. Model input data in the airGR format, as produced by ‘Create_Forcing_Inputs‘. Must include columns: DatesR, P_1 to P_n, E_1 to E_n, and Q (used for calibration).

`Subbasins` SpatVector. Geometries of subbasins. Must include attributes "COMID" (unique subbasin ID) and "Region" (region name/code).

`RunIni` character. Simulation start date in the format "mm/yyyy".

`RunEnd` character. Simulation end date in the format "mm/yyyy".

`Parameters` data.frame. GR2M model parameters and correction factors per region. Must have columns: Region, X1, X2, fp, fe.

`TransferMatrix` matrix or dgCMatrix. Subbasin connectivity matrix defining downstream routing. Rows represent receiving (downstream) subbasins, and columns represent donor (upstream) subbasins. Row and column names must match ‘COMID‘. For large river networks (thousands of subbasins), a sparse matrix is strongly recommended for memory efficiency.

`Outlet` character. Outlet subbasin identifier as a COMID code present in ‘Subbasins‘. Required for calibration.

`Parameters.Min` numeric vector of length 4. Lower bounds for optimization in the following order: c(X1, X2, fp, fe).

`Parameters.Max` numeric vector of length 4. Upper bounds for optimization in the same order as ‘Parameters.Min‘.

`Optimization` character. Objective function to optimize. Available options include:

- **"OF1"** Kling-Gupta Efficiency (KGE).
- **"OF2"** Nash-Sutcliffe Efficiency (NSE).
- **"OF3"** NSE applied to log-transformed flows (logNSE, Pushpalatha 2012).
- **"OF4"** Root Mean Square Error (RMSE).
- **"OF5"** Percent Bias (PBIAS).

"OF6" Bias in flow duration curves (FDC Bias).

"OF7" Pearson correlation coefficient (r).

"OF8" Composite objective combining KGE, NSE, and RMSE with weights (w1, w2, w3).

"OF9" Composite objective combining KGE, logNSE, and RMSE with weights.

"OF10" Composite objective combining KGE.km, KGE.lf, and RMSE with weights.

`Max.Functions` integer. Maximum number of function evaluations in the optimization. Default is 1000.

`WarmUp` integer. Number of months discarded as warm-up period when computing the objective function. If NULL (default), no warm-up is applied.

`No.Optim` character vector (optional). Region names to exclude from the optimization (parameters will be kept fixed).

`w1` numeric. Weight for the first component in composite objective functions (OF8, OF9, OF10). Default is 0.6.

`w2` numeric. Weight for the second component in composite objective functions. Default is 0.3.

`w3` numeric. Weight for the third component in composite objective functions. Default is 0.2.

Value

A list with the following components:

Parameters data.frame. Optimized parameter set per calibration region. Always includes all regions present in 'Subbasins'; regions excluded from optimization via 'No.Optim' retain their original parameter values. Columns are: Region, X1, X2, fp, fe.

OF numeric. Final value of the selected objective function corresponding to the best parameter set found by the optimization.

References

Llauca H, Lavado-Casimiro W, Montesinos C, Santini W, Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Define initial parameters for each region
param_init <- data.frame(
  Region = unique(cat$Region),
  X1  = 500,  # Production store capacity [mm]
  X2  = 1.5,  # Groundwater exchange coefficient
  fp  = 1.0,  # Precipitation correction factor
  fe  = 1.0   # Evapotranspiration correction factor
)

# Calibrate parameters using OF10 as objective function
result <- Optim_GR2MSemiDistr(
  Data = model_inputs,
```

```
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2005",
  TransferMatrix = matrixT,
  Outlet = '8', # basin outlet comid
  Parameters = param_init,
  Optimization = "OF10"
)

# Extract results
best_params <- result$Parameters
final_score <- result$OF

# Plot observed vs simulated discharge at the outlet
model <- Run_GR2MSemiDistr(
  Data = model_inputs,
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2005",
  TransferMatrix = matrixT,
  Outlet = '8', # basin outlet comid
  Parameters = best_params,
)

  par(mfrow = c(1, 1))
  plot(model$Dates, model$SINK$sim, type = "l", col = "blue", lwd = 1.5,
       xlab = "Date", ylab = "Discharge [m³/s]",
       main = "Simulated vs Observed Discharge (Outlet)")
    lines(model$Dates, model$SINK$obs, col = "red", lty = 2, lwd = 1.5)
    legend("topright", legend = c("Simulated", "Observed"),
           col = c("blue", "red"), lty = c(1, 2), lwd = 2, bty = "n")
```

`Run_GR2MSemiDistr` *Run the GR2M semi-distributed model with river network routing*

Description

Executes the monthly GR2M water balance model across multiple subbasins in a semi-distributed configuration. Region-specific parameters and correction factors are applied, and discharges are routed through the subbasin connectivity defined in a transfer matrix. The function supports a warm-up period, outlet extraction, saving results to disk, and an update mode for appending new results.

Usage

```
Run_GR2MSemiDistr(
  Data,
  Subbasins,
  RunIni,
  RunEnd,
  Parameters,
  TransferMatrix,
  Outlet = NULL,
```

```
  StatesIni = NULL,
  Save = FALSE,
  Update = FALSE
)
```

Arguments

`Data` data.frame. Model input data in the airGR format, as produced by 'Create_Forcing_Inputs'. Must include columns: DatesR, P_1 to P_n, E_1 to E_n, and optionally Q (observed discharge, used if available).

`Subbasins` SpatVector. Geometries of subbasins. Must include attributes "COMID" (unique subbasin ID) and "Region" (region name/code).

`RunIni` character. Simulation start date in the format "mm/yyyy".

`RunEnd` character. Simulation end date in the format "mm/yyyy".

`Parameters` data.frame. GR2M model parameters and correction factors per region. Must include columns: Region, X1, X2, fp, fe.

`TransferMatrix` matrix or dgCMatrix. Subbasin connectivity matrix defining downstream routing. Rows represent receiving (downstream) subbasins, and columns represent donor (upstream) subbasins. Row and column names must match 'COMID'. The matrix is internally reordered to align with input forcings. For large river networks (thousands of subbasins), a sparse matrix is strongly recommended for memory efficiency.

`Outlet` character (optional). COMID of the outlet subbasin to extract routed discharge. If provided, the simulated outlet discharge ('qsim') is returned, and compared with observed discharge ('Q') if available in 'Data'.

`StatesIni` list (optional). Initial state variables for GR2M subbasins. If 'NULL', default initial conditions are used.

`Save` logical. If 'TRUE', simulation results are saved in tab-separated '.txt' files inside the './Outputs' directory. Default is 'FALSE'.

`Update` logical. If 'TRUE', the function processes only the last available months of the input data and appends results to existing output files (if present). If the input contains only one month, an artificial second month is added internally to satisfy airGR's minimum timestep requirement, and this artificial record is discarded after the run. Default is 'FALSE'.

Details

In 'Update = TRUE' mode, the function processes the most recent months of the input series and appends them to existing outputs. If the input contains only a single month, an artificial second month (with P = 100 mm and E = 100 mm) is temporarily added to satisfy airGR's minimum timestep requirement. This artificial record is removed after the run, ensuring that only the real months are kept in the results and output files.

The transfer matrix describes subbasin connectivity: rows correspond to donor subbasins and columns to receivers. It can be supplied as a 'data.frame', a base R 'matrix', or a sparse 'dgCMatrix'. For applications with many subbasins (e.g., >10,000), a sparse representation is strongly advised to reduce memory usage and improve speed.

Value

A list with the following elements:

Dates Date vector. Simulation period (monthly).

COMID Character vector. Identifiers of the simulated subbasins.

PR matrix. Precipitation inputs per subbasin (mm). Columns correspond to subbasins.

AE matrix. Actual evapotranspiration per subbasin (mm).

SM matrix. Soil moisture storage per subbasin (mm).

PC matrix. Percolation per subbasin (mm).

RU matrix. Runoff generated per subbasin (mm).

QS matrix. Local discharge per subbasin (m^3/s), obtained from runoff conversion by area.

QR matrix. Routed discharge per subbasin (m^3/s), after applying the connectivity matrix.

SINK data.frame (optional). Simulated discharge at the outlet subbasin. Contains column 'sim' (simulated) and, if available, column 'obs' (observed). Returned only if 'Outlet' is provided.

StatesEnd list. Final GR2M states for each subbasin, as returned by 'airGR::RunModel_GR2M'.

References

Perrin C., Michel C., Andréassian V. (2003). Improvement of a parsimonious model for streamflow simulation. *Journal of Hydrology*, 279(1–4), 275–289. doi:10.1016/S00221694(03)002257

Llauca H., Lavado-Casimiro W., Montesinos C., Santini W., Rau P. (2021). PISCO_HyM_GR2M: A Model of Monthly Water Balance in Peru (1981–2020). *Water*, 13(8), 1048. doi:10.3390/w13081048

Examples

```
library(GR2MSemiDistr)

# Parameters (not calibrated)
parameters <- data.frame(
    Region = unique(cat$Region),
    X1  = 500,  # Production store capacity [mm]
    X2  = 1.5,  # Groundwater exchange coefficient
    fp  = 1.0,  # Precipitation correction factor
    fe  = 1.0   # Evapotranspiration correction factor
    )

# Run model with given parameters
model <- Run_GR2MSemiDistr(
  Data = model_inputs,
  Subbasins = cat,
  RunIni = "01/1981",
  RunEnd = "12/2016",
  Parameters = parameters,
  TransferMatrix = matrixT,
  Outlet = '8'
)

# Shown results for a target COMID
target_comid <- "10"
idx <- which(model$COMID == target_comid)

par(mfrow = c(3, 2),
    mar = c(4, 4, 3, 1),
    cex.main = 0.9,
```

```
    cex.lab = 0.8,
    cex.axis = 0.7)

# Precipitation
plot(model$Dates, model$PR[, idx], type = "h", lwd = 2, col = "dodgerblue",
     main = sprintf("Precipitation (PR) - COMID %s", target_comid),
     xlab = "Date", ylab = "mm/month")

# Actual evapotranspiration
plot(model$Dates, model$AE[, idx], type = "h", lwd = 2, col = "orange",
     main = sprintf("Actual Evapotranspiration (AE) - COMID %s", target_comid),
     xlab = "Date", ylab = "mm/month")

# Soil moisture storage
plot(model$Dates, model$SM[, idx], type = "h", lwd = 2, col = "green",
     main = sprintf("Soil Moisture (SM) - COMID %s", target_comid),
     xlab = "Date", ylab = "mm")

# Percolation
plot(model$Dates, model$PC[, idx], type = "h", lwd = 2, col = "purple",
     main = sprintf("Percolation (PC) - COMID %s", target_comid),
     xlab = "Date", ylab = "mm/month")

# Runoff
plot(model$Dates, model$RU[, idx], type = "h", lwd = 2, col = "red",
     main = sprintf("Runoff (RU) - COMID %s", target_comid),
     xlab = "Date", ylab = "mm/month")

# Routed discharge
plot(model$Dates, model$QR[, idx], type = "l", lwd = 1.5, col = "blue",
     main = sprintf("Routed Simulated Discharge (QR) - COMID %s", target_comid),
     xlab = "Date", ylab = "m³/s")

# Compare simulated vs observed discharge at outlet
par(mfrow = c(1, 1))
plot(model$Dates, model$SINK$sim, type = "l", col = "blue", lwd = 1.5,
     xlab = "Date", ylab = "Discharge [m³/s]",
     main = "Simulated vs Observed Discharge (Outlet)")
  lines(model$Dates, model$SINK$obs, col = "red", lty = 2, lwd = 1.5)
  legend("topright", legend = c("Simulated", "Observed"),
         col = c("blue", "red"), lty = c(1, 2), lwd = 2, bty = "n")
```

Index